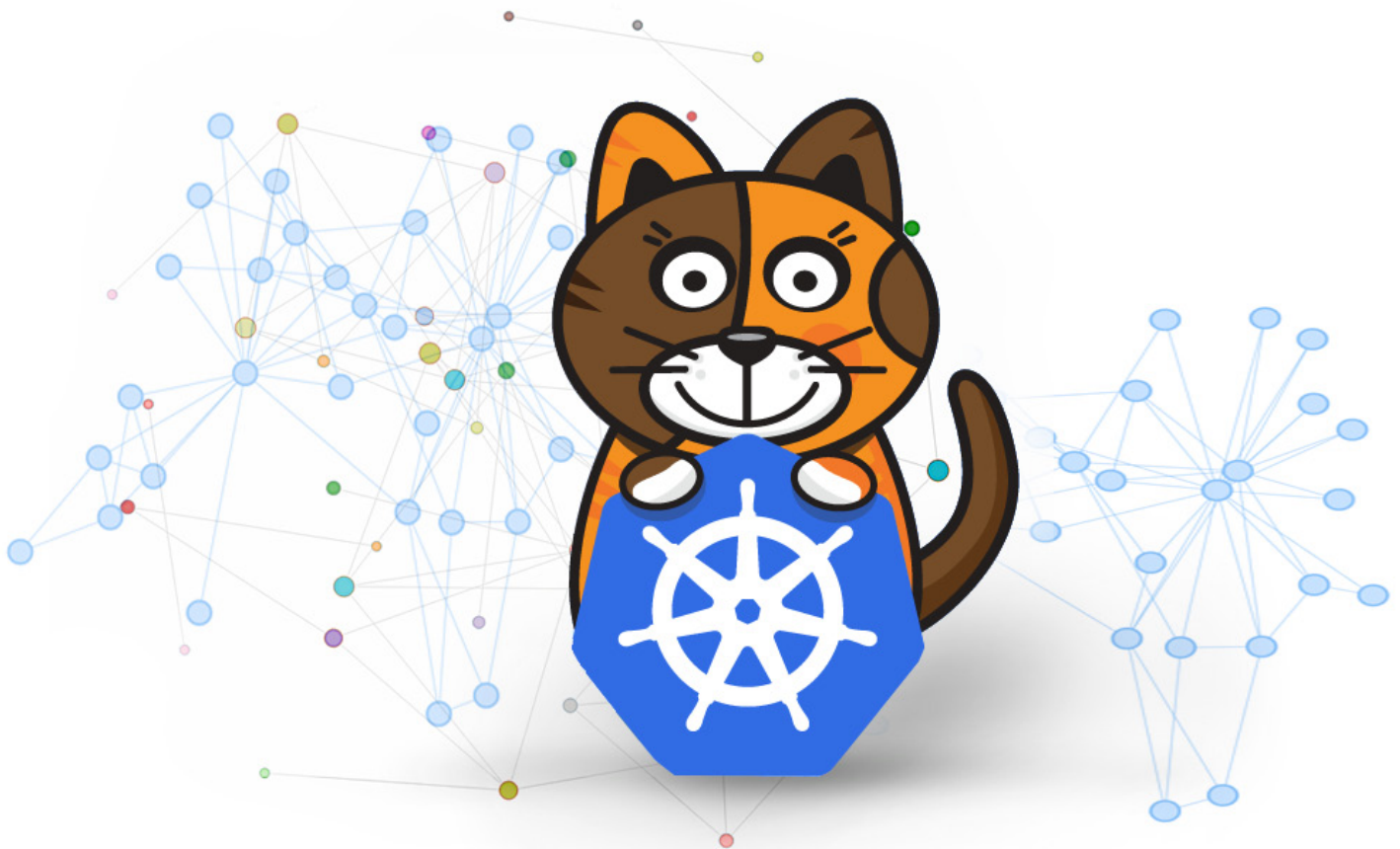


Enhancing Kubernetes with Calico Network Policies



This article gives a demo of how a container network interface (CNI) and, more specifically, Calico works and extends the Kubernetes platforms.

Kubernetes is designed to support modular cloud-native applications. The platform itself is modular and flexible. It allows users to integrate plugins, extensions, add-ons, services and interfaces to extend the core functionality of the platforms. One can classify interfaces broadly into three categories — container network interface (CNI), container runtime interface (CRI), and container storage interface (CSI).

Kubernetes and Calico: The background

Networking is an integral part of the Kubernetes platform. In a typical Kubernetes cluster, we are trying to solve four problems in general:

1. Highly coupled container to container communication
2. Pod to pod communication
3. Pod to service communication
4. External world to service communication

A container network interface or CNI tries to solve these problems by implementing a networking model in Kubernetes. It supports plugin-based functionality to simplify and extend networking in Kubernetes. The main purpose of CNI is to increase control over the growing network while minimising the overhead of managing the network manually.

There are multiple CNI plugins available, which can be added to your Kubernetes cluster; for example, Calico, Flannel, WeaveNet, etc. You can find out more details about